



**DEPARTMENT OF INFORMATION
& COMMUNICATION TECHNOLOGY**



DIPLOMA IN INFORMATION TECHNOLOGY

**PROPOSAL
FYP INVENTORY SYSTEM**

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SESSION I:2026/2027

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1.0 INTRODUCTION

The rapid advancement of information technology has significantly transformed the way educational institutions manage academic information and documentation. Many organizations have shifted from conventional record management to digital systems to improve efficiency, accessibility, and data security. In higher education institutions, digital information systems play an important role in organizing academic records and supporting knowledge sharing among students, lecturers, and administrators. A well-developed information system not only simplifies administrative tasks but also provides a centralized platform for managing valuable academic resources.

At **Politeknik Besut Terengganu (POLIBESUT)**, the Final Year Project (FYP) is a compulsory course undertaken by Semester 5 students in the **Department of Information and Communication Technology (JTMK)** and the **Department of Visual Communication and Design (JRKV)**. Throughout the project, students are required to produce various documents and project deliverables, including proposals, final reports, presentation slides, project images, and supporting materials. These documents represent valuable academic references for future students and lecturers. However, the current method of managing these project records is limited and does not fully support long-term documentation and knowledge sharing.

Based on the preliminary investigation conducted with the Final Year Project Coordinator, the existing **iSOLMS** system mainly focuses on the registration and management of Final Year Project activities such as student registration, supervisor assignment, and project summaries. Although the system supports the administrative process, it does not function as a comprehensive repository for storing complete project documentation. Students are only able to view limited project information, while important materials such as full reports, proposals, presentation slides, project images, Gantt charts, and other supporting documents are not systematically organized for future reference.

Therefore, this project proposes the development of a web-based **FYP Inventory System** that serves as a centralized digital repository for Final Year Project documentation. The proposed system enables Semester 5 students to upload complete project information, while lecturers, coordinators, and other students can browse previous projects as academic references. To protect intellectual property, all uploaded documents are provided in **view-only mode**, preventing users from downloading, copying, or modifying the original files. In addition, the system categorizes projects into various fields such as **Website, Mobile Application, Desktop Application, Game, Information System, Craft, and Batik**, allowing users to search projects more efficiently according to their interests.

The proposed system will be developed as a web-based application using modern web development technologies, including **HTML, CSS, JavaScript, PHP, and MySQL**. These technologies provide a reliable platform for developing a secure, responsive, and user-friendly application capable of managing large volumes of Final Year Project records. It is expected that the implementation of the FYP Inventory System will improve project documentation management, facilitate knowledge sharing, and become a sustainable digital repository for future academic use within JTMK and JRKV.

2.0 PROBLEM STATEMENT

The current **iSOLMS** system only provides basic information about Final Year Projects, such as the project title and a brief summary. Important project documentation, including the proposal, final report, presentation slides, project images, and Gantt chart, is not available within the system. As a result, students and lecturers are unable to access comprehensive project information for academic reference. This limitation reduces the usefulness of the existing system as a repository for previous Final Year Projects.

In addition, there is no centralized platform that systematically stores all Final Year Project documentation. Project files are often kept separately by students or supervisors, making them difficult to retrieve once the project has been completed. This situation increases the risk of data loss and makes the management of project records less efficient. Furthermore, the Final Year Project Coordinator faces difficulties in maintaining a complete inventory of projects from different semesters.

Another issue is that students from lower semesters have limited access to previous Final Year Projects that could serve as references for generating ideas and understanding project development. Since the existing system does not provide complete documentation or an efficient search function, students spend more time searching for relevant references through lecturers or former students. Consequently, the lack of an organized digital repository affects knowledge sharing and reduces opportunities for students to learn from previous projects.

Therefore, the development of the **FYP Inventory System** is proposed to overcome these issues by providing a centralized, web-based platform that enables systematic storage, efficient management, and easy access to complete Final Year Project documentation. The system also includes search and categorization features, allowing students, lecturers, and coordinators to retrieve project information quickly while ensuring that uploaded documents remain protected through view-only access.

3.0 OBJECTIVE

The following are the objectives of the proposed **FYP Inventory System**:

- i. **To identify the current Final Year Project documentation and inventory management process** at the Department of Information and Communication Technology (JTMK) and the Department of Visual Communication and Design (JRKV).
- ii. **To design and develop a web-based Final Year Project Inventory System** that enables students to upload, manage, and organize Final Year Project documentation in a centralized digital repository.
- iii. **To evaluate the usability and functionality of the proposed Final Year Project Inventory System** through user evaluation using questionnaires and system testing.

4.0 SCOPE

4.1 User Scope

4.1.1 Student

- Students can login to the system.
- Students can register their Final Year Project (FYP).
- Students can upload all the necessary documents, which include the project proposal, final report, poster, presentation slides, Gantt chart, and system screenshots.
- Students can check the status of their project, e.g., *In Progress*, *Approved* or *Rejected*.
- Students can view their project marks and assessment.
- Students can search for previously done Final Year Projects for reference.

4.1.2 Supervisor

- Supervisors can keep track of the progress of students that they supervise.
- Supervisors can download the projects done by students for assessing them.
- Supervisors can assess projects by giving marks and commenting on them according to the assessment rubric.
- Supervisors can approve projects before submission to the Head of Department (HoD).

4.1.3 Administrator (Admin)

- The administrator has the ability to handle project registrations and student group registrations.
- The administrator can create, update, and delete assessment rubrics.
- The administrator can post announcements regarding the Final Year Project.
- The administrator is responsible for analyzing the project marks and generating project rankings.

4.1.4 Head of Department (HoD)/ Panel

- The Head of Department (HoD) can moderate the assessment marks submitted by the supervisor.
- The Head of Department can approve the final marks for the project.
- The Head of Department can generate Final Year Project ranking reports.
- The panel members can assess special projects and give marks if needed.

4.2 System Scope

4.2.1

The system will have a login functionality for all users which include students, supervisors, administrator (Admin) and Head of Department (HoD)/panel members.

4.2.2

The system enables the students to upload their Final Year Project (FYP) documents and store all documents uploaded into the database.

4.2.3

The system will enable the supervisors to download their students' project documents, award marks, comment and approve projects.

4.2.4

The system will enable the administrator to manage rubric of assessments, post announcements, analyse project marks and track project progress.

4.2.5

The system will enable the Head of Department (HoD) and the panel members to moderate project marks, rank projects and approve final project results.

4.2.6

The system will automatically generate performance reports for Final Year Projects and these include marks awarded, ranking of projects and the project status.

4.2.7

The system will be able to perform search and will allow the students and lecturers to retrieve previously submitted Final Year Projects.

5.0 PROJECT SIGNIFICANCE

The development of the **FYP Inventory System** provides various benefits to students, lecturers, the Final Year Project Coordinator, and the departments involved. The system is expected to improve the management of Final Year Project documentation while supporting knowledge sharing and efficient information retrieval. The following are the importance and significance of the system :

- 5.1 The proposed system provides students with a centralized platform to upload and manage their Final Year Project documentation. Students can also search and view previous projects based on categories, allowing them to gain ideas and references for their own projects. By having access to complete project documentation in a view-only environment, students are encouraged to learn from previous work while maintaining academic integrity.
- 5.2 The system enables lecturers to monitor students' Final Year Project documentation more efficiently. Lecturers can review uploaded documents, provide comments, and record project evaluation marks through a single platform. This reduces the need for manual document management and improves communication between lecturers and students throughout the project development process.
- 5.3 The FYP Inventory System assists the Final Year Project Coordinator in managing project records from different semesters in a more systematic manner. The coordinator can verify uploaded projects, organize project information according to category and semester, generate reports, and maintain a centralized inventory of Final Year Projects. This improves administrative efficiency and simplifies the process of retrieving project information whenever required.
- 5.4 The implementation of the proposed system benefits both the **Department of Information and Communication Technology (JTMK)** and the **Department of Visual Communication and Design (JRKV)** by creating a sustainable digital repository for Final Year Projects. The system preserves valuable academic resources, promotes knowledge sharing among students and lecturers, and supports continuous improvement in project quality. Furthermore, it strengthens the departments' documentation management practices by ensuring that project information is systematically stored, easily accessible, and well organized for future academic and reference purposes.

6.0 LITERATURE REVIEW

A literature review is conducted to study existing systems that provide similar functions to the proposed **FYP Inventory System**. The purpose of this review is to identify the strengths and limitations of current systems and to determine how the proposed system can overcome existing problems. Three existing systems have been selected for comparison, namely **iSOLMS**, **DSpace Institutional Repository**, and **Google Drive**.

6.1 iSOLMS

The **Integrated Student Outcome Management System (iSOLMS)** is a web-based learning management and project administration system used by Malaysian Polytechnics. The system is developed to support **Project Based Learning (PBL)** and assists in managing several academic activities, including Final Year Project (FYP) administration. Students who are registered for the Final Year Project course are granted access to the system to perform project-related activities.

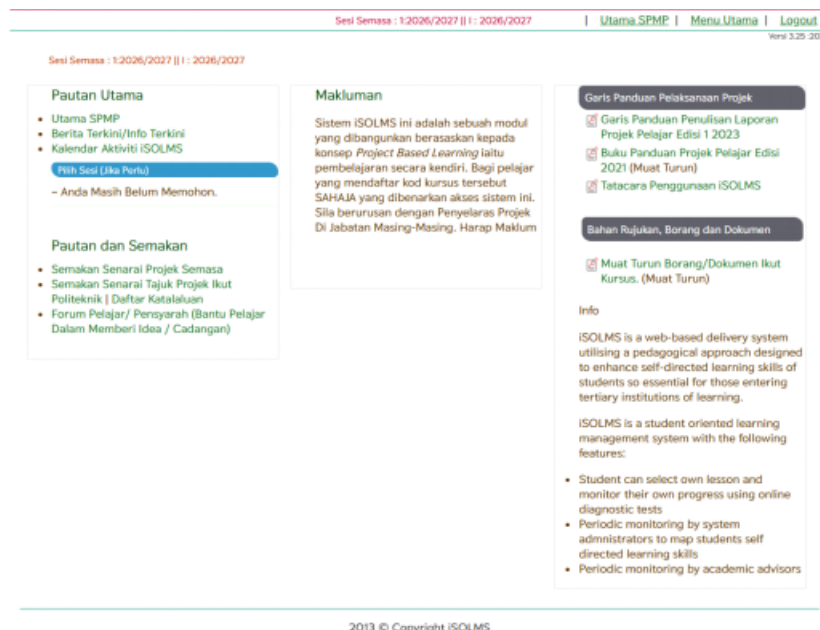


Figure 6.1: iSOLMS Homepage
(Source: *Student Oriented Learning Management System (iSOLMS)*,
<https://spmp.polibesut.edu.my/isolms/index.jsp>)



Figure 6.2: Session and Department Selection Page for Current Final Year Projects in iSOLMS

According to the current implementation, iSOLMS provides several functions such as project registration, project title listing, project guidelines, downloadable forms and documents, and project information. Students can also check the current project list and search for previously registered project titles through the system. In addition, lecturers and coordinators are able to monitor students' project progress using the available modules.

Advantages	Disadvantages
· Provides an online platform for Final Year Project administration. · Allows students to register their Final Year Project. · Displays the list of current and previous project titles. · Provides project guidelines, manuals, and downloadable documents. · Enables lecturers and coordinators to monitor project activities. · Accessible through a web browser.	· Only basic project information is available, such as project titles and summaries. · Does not provide a centralized repository for complete project documentation. · Students cannot upload complete project files such as reports, proposals, presentation slides, Gantt charts, or project images for future reference. · Limited search functionality based on project categories. · Does not function as a digital archive for previous Final Year Projects.

The proposed **FYP Inventory System** extends the functionality of iSOLMS by providing a comprehensive digital repository that stores complete Final Year Project documentation. Unlike iSOLMS, the proposed system allows students to upload reports, proposals, presentation slides, Gantt charts, project images, and other supporting documents in a **view-only** format. The system also introduces project categorization and an advanced search feature, making it easier for students and lecturers to locate previous projects based on categories such as Website, Mobile Application, Game, Information System, Craft, and Batik.

6.2 DSpace Institutional Repository

DSpace is an open-source digital repository platform widely used by universities and research institutions to preserve and manage academic publications, theses, dissertations, research papers, and institutional documents. The system enables users to search and access digital collections through an organized repository.



Figure 6.3: DSpace About Page
(Source: DSpace, <https://dspace.org/about/>)

The page introduces DSpace as an open-source repository software used by academic institutions, non-profit organizations, and commercial organizations to build digital repositories. It also provides general information about the purpose of DSpace and its role in preserving and managing digital content.

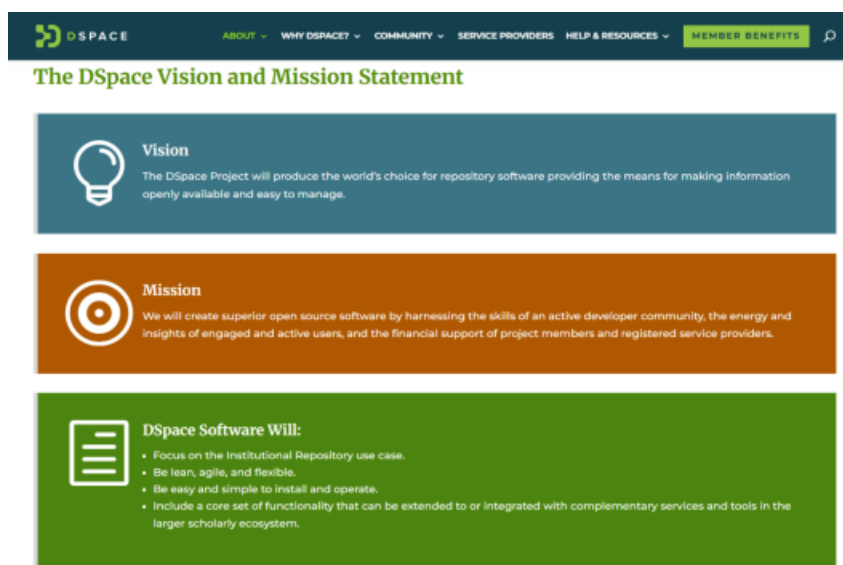


Figure 6.4: DSpace Vision and Mission
(Source: DSpace, <https://dspace.org/about/vision-and-mission/>)

The page highlights DSpace's commitment to providing an open-source repository platform that supports the long-term storage, management, preservation, and accessibility of digital content. It also emphasizes continuous development through community collaboration and flexible repository solutions for educational and research institutions.

Advantages	Disadvantages
· Supports digital preservation. · Open-source platform. · Powerful search and metadata management. · Suitable for academic repositories. · Supports long-term document storage.	· Not specifically designed for Final Year Project management. · Requires technical knowledge for installation and maintenance. · Does not include lecturer supervision or project evaluation modules. · More suitable for institutional repositories than classroom project management.

6.3 Google Drive

Google Drive is a cloud-based storage service developed by Google that allows users to upload, store, organize, and share digital files online. Many educational institutions use Google Drive to store academic documents due to its ease of use and accessibility.

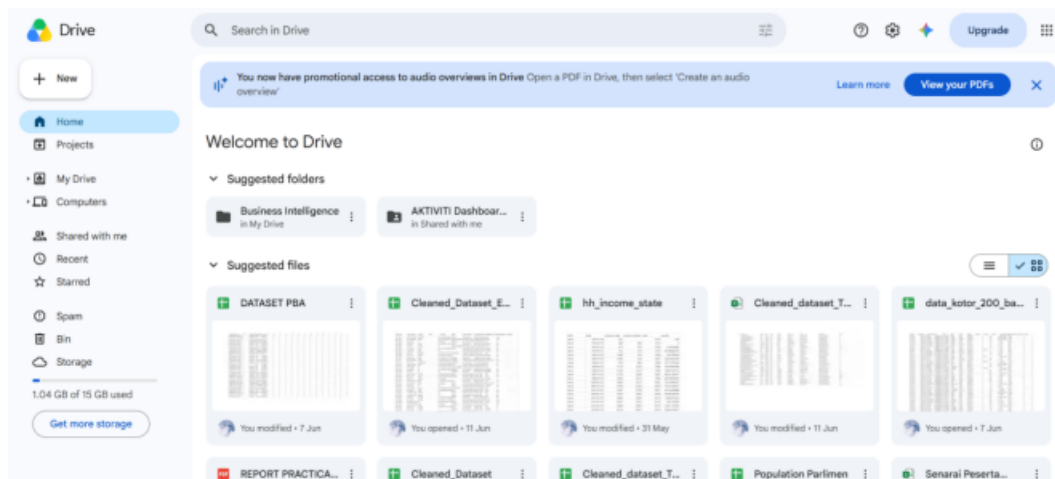


Figure 6.5: Google Drive Home Interface

(Source: Google Workspace, <https://workspace.google.com/products/drive/>)

The figure above shows the main interface of Google Drive after a user log into their account. The homepage provides quick access to recently used files, folders, and navigation menus such as **My Drive**, **Computers**, **Shared with me**, **Recent**, and **Starred**. Users can also upload files, create folders, and organize documents using the available storage space. The search bar at the top enables users to locate files efficiently based on file names or keywords.

Google Drive serves as a cloud storage platform that allows users to store and access digital files from anywhere with an internet connection. However, although it supports file storage and sharing, it does not provide specialized features for managing academic Final Year Project (FYP) records such as project categorization, supervisor monitoring, student assessment, or project inventory management.

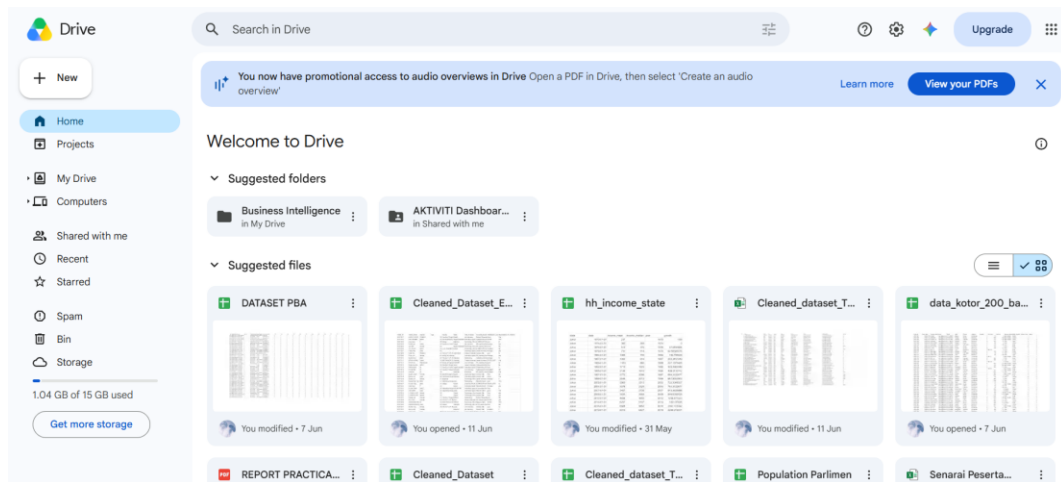


Figure 6.6: Google Drive Shared with Me Interface

(Source: Google Workspace, <https://workspace.google.com/products/drive/>)

The figure above illustrates the **Shared with me** section in Google Drive, where users can access files and folders shared by other users. Documents are displayed in a grid layout together with file previews, making it easy for users to identify and open shared resources. The interface also provides filtering options based on file type, people, modification date, and source.

Although this feature allows collaborative file sharing, it lacks dedicated functionalities required for FYP management. There is no centralized project inventory, project information page, category-based searching, supervisor evaluation, or project documentation management. Therefore, Google Drive is suitable as a general cloud storage platform but is insufficient as a comprehensive repository system for managing Final Year Project information.

Advantages	Disadvantages
• Easy to upload and share files. • Accessible from various devices. • Supports cloud storage. • Free storage for basic users.	• Not specifically designed for Final Year Project management. • No project categorization according to academic requirements. • No lecturer evaluation or marking features. • Difficult to manage a large number of projects systematically.

Unlike Google Drive, the proposed **FYP Inventory System** is specifically developed to manage Final Year Project documentation. Besides storing files, the system also organizes projects according to semester and category, supports lecturer evaluation, and provides a structured digital inventory for academic references.

7.0 METHODOLOGY

The chosen system development model for the current project is the **Agile Model**. Agile is a system development approach where iterations called sprints are conducted during the development process and where collaboration between developers and users is done continuously. In addition, feedback on the system being developed is continuously obtained and used in order to improve the system in every subsequent iteration.

The **Agile Model** was chosen due to the fact that the **Inventory FYP** system contains several important modules such as login module, document submission module, rubric-based assessment module, and ranking module. Thus, using the Agile approach, it becomes possible to develop, test and improve the modules incrementally without interfering with the whole system development process.



Figure 7.1: Flow chart depicting Agile Model.

Phase 1: Planning

In this phase, there will be collaboration between the project team members, students, supervisors, administration, and the Head of Department (HOD) in order to find out the key requirements of the system. The key product backlog in this phase will contain various modules of the system, such as login module, document submission module, rubric-based evaluation module, and project ranking module.

Phase 2: Design

In this phase, the initial design of the system will be developed by the project team, which includes UI design, database schema, and process flow. This design will be flexible and may change according to the user suggestions throughout the development process. The key output of this phase will be the initial prototype of Inventory FYP system.

Phase 3: Development

The development of the system will be incremental and will be completed through several sprints. For instance, the first sprint will focus on login module, the second sprint will be for document submission module, and the third sprint will be for the supervisor evaluation module.

Phase 4: Testing

In this phase, each sprint undergoes tests conducted by both students and supervisors to ensure the proper operation of the implemented features. It involves the login process, document upload, rubric based evaluation, and ranking of projects. Output of this phase includes testing reports and feedback.

Phase 5: Review and Feedback

After the conclusion of each sprint, reviews with users are done to assess the efficiency of the implemented features in the system. Feedback gathered is useful in making improvements to the system in the following sprints. Output of this phase includes an improved system that satisfies users' demands.

Phase 6: Implementation

Deployment of the system for use by students, supervisors, administrators, and the Head of Department (HOD) takes place in this phase. Output of this phase includes a complete system.

8.0 REFERENCE

DSpace. (n.d.). *About DSpace*. DSpace. Retrieved July 6, 2026, from <https://dspace.org/about/>

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9.0 GANTT CHART

The Gantt Chart illustrates the project schedule and timeline for the development of the **FYP Inventory System**. It outlines the sequence of activities from the planning phase until the deployment phase within a period of fourteen (14) weeks. The chart is used to monitor project progress and ensure that every development phase is completed according to the planned schedule.

Table 9.1: Gantt Chart for FYP Inventory System Development

Based on the Gantt

Activities/Phases	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
Planning														
Design														
Development Sprint 1														
Development Sprint 2														
Development Sprint 3														
Testing														
Review & Feedback														
Deployment														

Chart, the project begins with the planning and system design phases, followed by three development sprints focusing on different system modules. After the development process is completed, the system undergoes testing, review, and deployment. This schedule ensures that each phase is carried out systematically and within the proposed project duration.

10.0 COST PLANNING

Cost planning is an important aspect in the development of the proposed FYP Inventory System. It provides an estimation of the resources required throughout the project development process. Although most of the software used in this project is open-source and free of charge, several costs such as hardware usage, internet access, printing, transportation, and miscellaneous expenses are considered to ensure the successful completion of the project.

Table 10.1: Estimated Cost for FYP Inventory System Development

No.	Item	Quantity	Estimated Cost (RM)
1.	Laptop (Personal Device)	1	0.00*
2.	Internet Subscription	5 Months	150.00
3.	Visual Studio Code	1	0.00
4.	XAMPP	1	0.00
5.	MySQL	1	0.00
6.	Google Chrome	1	0.00
7.	Transportation	-	50.00
8.	Printing	-	30.00
9.	Stationery	-	20.00
10.	USB Flash Drive	1	30.00
Total			RM 280.00

*The laptop is an existing personal device owned by the project developer; therefore, no purchase cost is incurred.

The estimated development cost for the proposed FYP Inventory System is RM280.00. Since the project utilizes free and open-source software such as Visual Studio Code, XAMPP, and MySQL, the overall development cost is significantly reduced. Most of the estimated expenses are allocated to internet access, transportation for meetings and consultations, printing of project documents, and other miscellaneous costs incurred during the development process.

The proposed budget is considered reasonable and sufficient for developing the system while ensuring that all project activities can be completed successfully within the allocated time frame.

11.0 CONCLUSION

Concluding, the development of the Inventory FYP System using the Agile Methodology has managed to meet all the requirements of the final year project management in the faculty. This was accomplished by developing the system in various sprints, including the user login module, document upload module, rubrics-based marking system, and project ranking module among others. Every sprint was subjected to testing and improvement according to user suggestions, hence a flexible and easy-to-use system.

First of all, the system allows the students to upload their project documents and communicate easily with their supervisors. Secondly, the system helps in systematic management and evaluation of final-year projects by the administrators and Head of Department. Therefore, the project management has become more organized and efficient compared to the previously used manual system.

Thus, the Inventory FYP System has successfully met its objectives by developing a fully functional platform for the management of final-year projects. This project shows that the use of the Agile methodology is appropriate when developing systems that require improvements according to user feedback.

For future development, the system can be enhanced by incorporating several additional features. These include integration with the university's official portal to synchronize student data, the implementation of an analytics module to generate performance reports automatically, and the development of a mobile application to improve accessibility for both students and lecturers. In addition, a real-time notification feature could be introduced to remind students of document submission deadlines and to notify them of supervisor feedback promptly. With these enhancements, the Inventory FYP System has the potential to become a more comprehensive, modern, and efficient platform for managing final-year projects while meeting the evolving needs of higher education institutions.